HOME IT &

CYBERSECURITY LAB

Technical Portfolio

WINDOWS ADMINISTRATION | IDENTITY & ACCESS MANAGEMENT | NETWORKING
GLPI SERVICE DESK | WAZUH SECURITY MONITORING | TROUBLESHOOTING

FINAL INTEGRATED EDITION WITH SELECTED VISUAL EVIDENCE | OCTOBER 26, 2026

A practical, evidence-led lab environment for Windows and Linux administration, identity management, network troubleshooting, service-desk workflow, and endpoint security monitoring.

This portfolio documents work performed in a personal learning environment. Controlled test scenarios, measured outcomes, and verification limitations are identified so that configuration work is not confused with a demonstrated result.

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Ticket chronology note: GLPI dates and times are reproduced from the exported ticket records. The export did not identify the timezone for the stored timestamps.

Executive Summary

This portfolio documents a hands-on home IT and cybersecurity lab built around VMware Workstation Pro, Windows Server, Active Directory, a domain-joined Windows client, a Linux system hosting GLPI, and a Wazuh security-monitoring stack. The environment is used to practice configuration, identity administration, fault reproduction, troubleshooting, access validation, service recovery, ticket documentation, and review of endpoint security events.

The work demonstrates several connected technical disciplines: maintaining an isolated virtual lab network; administering a Windows domain; organizing users by department and assigning permissions through security groups; documenting support cases in GLPI; and using Wazuh to observe selected Windows security events. The Northstar Services exercise extended the domain into a fictional five-department organization with 25 user accounts, five departmental Global Security groups, departmental NTFS permissions and SMB shares, and user-based tests of allowed and denied access.

The GLPI register records 12 incidents with their requesters, affected systems, investigation steps, remediation, and observed outcomes. Most tickets demonstrate recovery or successful troubleshooting. Ticket 11 documents a firewall-policy investigation without configuration changes, and Ticket 12 is explicitly identified as technically unverified after the attempted Linux permission changes, despite its administrative closed status.

Portfolio principle Results are described at the level supported by the recorded evidence. A controlled failure is not presented as an unexpected production incident, an unsuccessful access attempt is not assigned an unrecorded error code, and a closed ticket is not automatically treated as proof of successful technical recovery.

1. Objectives and Approach

The lab was designed to provide practical experience across common IT support and entry-level cybersecurity activities while using a low-cost mix of available, evaluation, and open-source software where practical.

- Build and maintain an isolated multi-VM environment for Windows, Linux, directory services, service-desk software, and endpoint monitoring.
- Practice Active Directory user and group administration, account lifecycle operations, domain authentication, DNS, Group Policy, and group-based resource access.
- Investigate Windows client and network problems using observable state, logs, and repeatable tests instead of assuming that an application is the source of a connectivity failure.
- Use GLPI to preserve the symptom, impact, affected asset, investigation, corrective action, verification, and closure history for support incidents.
- Use Wazuh to distinguish a Windows event being present from a matching rule or alert being verified.
- Preserve technical checkpoints and document known limitations without overstating test coverage.

The project emphasizes evidence-based troubleshooting: observe the starting state, make a controlled change, test the expected outcome, and record any remaining uncertainty. The lab is a learning environment, not a claim of production enterprise deployment or a production security operations center.

2. Infrastructure and Network Architecture

2.1 Host and virtualization platform

Component	Recorded configuration
Host system	Mike-Lab - Dell OptiPlex 7070
Processor / memory / storage	Intel Core i7-9700 at 3.4 GHz; 32 GB DDR4; 1 TB SSD
Host operating system	Windows 11 Pro 25H2, build 26200.9457 (recorded)
Hypervisor	VMware Workstation Pro 25H2
VM storage root	C:\Lab\Virtual Machines

VMware displayed a Hyper-V / Device-Credential Guard and Windows Hypervisor Platform warning during setup. The lab was used with that host configuration; the warning was not confirmed as the cause of a specific failure.

2.2 Virtual networking

Network / component	Recorded addressing and purpose
VMnet1 - host-only	192.168.247.0/24. Host virtual adapter: 192.168.247.1. VMware DHCP server: 192.168.247.254 when enabled. This is the primary isolated lab segment.
VMnet8 - NAT	192.168.2.0/24 recorded as the NAT segment for optional outbound access.
Lab isolation	The home LAN is separate from the host-only lab segment. Bridging guests directly to the home LAN is not the default design.

All addresses below are historical recorded values, not a guarantee of current live configuration. They are included to explain the documented troubleshooting work and should be rechecked before future changes.

2.3 Virtual machine inventory

VM	Role and recorded details
LAB-DC01	Windows Server 2025 Evaluation; Active Directory Domain Services and DNS for ad.home.arpa; 2 vCPU, 4 GB RAM, 60 GB virtual disk; historical address 192.168.247.2.
LAB-CLIENT01	Domain-joined Windows client used for endpoint administration and security tests; Wazuh agent ID 002. Address 192.168.247.128/24 was observed after the October 20 DHCP recovery; an earlier record also lists 192.168.247.101.
LAB-WAZUH01	Wazuh manager, indexer, and dashboard; recorded address 192.168.247.20.
LAB-LNX01	Linux VM hosting GLPI and the Apache web service; historical address 192.168.247.10.

The two pre-existing Windows file shares on LAB-DC01 are Department (C:\LabShares\Department) and IT (C:\LabShares\IT). They were inspected during Northstar setup and left unchanged.

3. Active Directory, Identity and Access Management

Active Directory is the identity and organizational foundation of the lab. The domain is ad.home.arpa, with LAB-DC01 providing AD DS and DNS and LAB-CLIENT01 serving as the domain-joined Windows endpoint. The documented administration work includes domain setup, user and group administration, account state changes, Group Policy troubleshooting, and group-based file access.

3.1 HelpDeskLab administration

On October 25, 2026, Active Directory Users and Computers on LAB-DC01 was used to create and verify a small identity-lifecycle exercise:

- Created the HelpDeskLab organizational unit and the domain user alex.morgan within it.
- Administratively reset the account password; no password value is recorded.
- Created GG-HelpDeskLab-Test with Global scope and Security group type.
- Added alex.morgan to GG-HelpDeskLab-Test and verified membership from the group Members tab and the user Member Of tab.
- Removed alex.morgan from the group and verified it no longer appeared in Member Of.
- Disabled the account, verified its disabled state, re-enabled it, and verified the enabled state.
- Set the Description attribute to "Help Desk Lab Test Account" and reopened the account properties to verify the saved value.

Final state recorded alex.morgan is enabled and is not a member of GG-HelpDeskLab-Test. These were Active Directory administration checks; no Wazuh detection is claimed for these specific actions.

3.2 Northstar Services directory structure

On October 26, 2026, the existing ad.home.arpa domain was extended with a fictional organization, Northstar Services. The structure was created manually in Active Directory Users and Computers to provide direct practice with organizational units, user creation, group objects, and membership management. The existing HelpDeskLab OU and unrelated lab accounts were left untouched.

Organizational layout ad.home.arpa > Northstar Services > IT | Human Resources | Finance | Operations | Customer Support | Groups

The five departmental OUs contain five fictional employee accounts each. The Groups OU contains five Global Security groups, one for each department:

Department OU	Global Security group	Purpose
IT	GG-NS-IT	Northstar IT department access
Human Resources	GG-NS-HR	Northstar HR department access
Finance	GG-NS-Finance	Northstar Finance department access
Operations	GG-NS-Operations	Northstar Operations department access
Customer Support	GG-NS-CustomerSupport	Northstar Customer Support department access

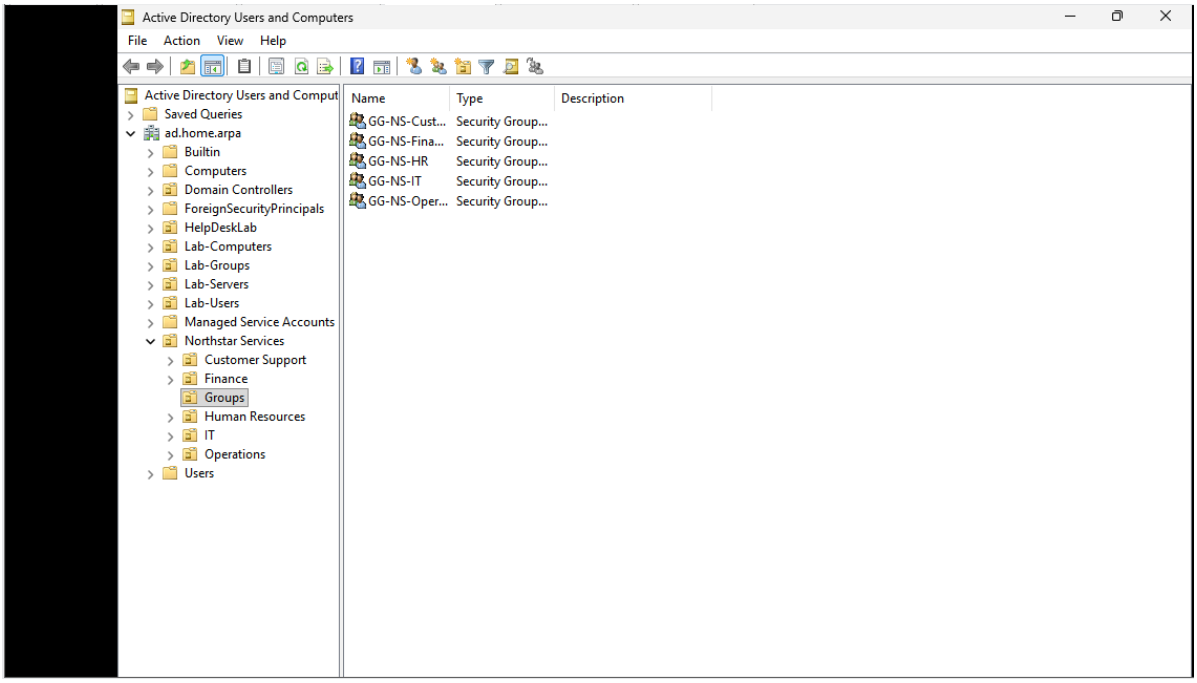


Figure 1. Active Directory Users and Computers showing the Northstar Services OU hierarchy, with the Groups OU selected and departmental security groups listed.

3.3 Northstar account inventory

All accounts below were manually created in Active Directory Users and Computers and placed directly in the appropriate department OU. The first.last logon-name convention was used. Names are fictional; no passwords are included.

Department	Fictional employee	Logon name	Group membership
IT	Jordan Blake	jordan.blake	GG-NS-IT
IT	Morgan Ellis	morgan.ellis	GG-NS-IT
IT	Casey Rivera	casey.rivera	GG-NS-IT
IT	Avery Brooks	avery.brooks	GG-NS-IT
IT	Taylor Nguyen	taylor.nguyen	GG-NS-IT
Human Resources	Jamie Foster	jamie.foster	GG-NS-HR
Human Resources	Riley Bennett	riley.bennett	GG-NS-HR
Human Resources	Cameron Hayes	cameron.hayes	GG-NS-HR
Human Resources	Sydney Parker	sydney.parker	GG-NS-HR
Human Resources	Drew Collins	drew.collins	GG-NS-HR
Finance	Alex Turner	alex.turner	GG-NS-Finance
Finance	Bailey Morgan	bailey.morgan	GG-NS-Finance
Finance	Devon Reed	devon.reed	GG-NS-Finance
Finance	Quinn Murphy	quinn.murphy	GG-NS-Finance
Finance	Harper James	harper.james	GG-NS-Finance
Operations	Logan Price	logan.price	GG-NS-Operations
Operations	Parker Simmons	parker.simmons	GG-NS-Operations
Operations	Reese Coleman	reese.coleman	GG-NS-Operations
Operations	Jordan Wells	jordan.wells	GG-NS-Operations
Operations	Skyler Bennett	skyler.bennett	GG-NS-Operations
Customer Support	Emerson Ward	emerson.ward	GG-NS-CustomerSupport
Customer Support	Finley Ross	finley.ross	GG-NS-CustomerSupport
Customer Support	Rowan Diaz	rowan.diaz	GG-NS-CustomerSupport
Customer Support	Blake Cooper	blake.cooper	GG-NS-CustomerSupport
Customer Support	Kendall Scott	kendall.scott	GG-NS-CustomerSupport

3.4 Provisioning method and account options

The user accounts were created manually through the graphical interface rather than by a bulk-provisioning script. This provided direct practice creating accounts and managing group membership. Jordan Blake was initially added using the user Properties > Member Of workflow; the remaining IT users and the departmental users were added through each security group Properties > Members tab. Each department group was checked to confirm five members.

For this disposable lab population, "User must change password at next logon" was unchecked to make repeat exploratory sign-in practical. "User cannot change password," "Password never expires," and "Account is disabled" were left unchecked. These choices are documented as lab convenience settings, not as a production password policy or a recommended configuration for live employee accounts.

3.5 Northstar departmental folders and NTFS permissions

A separate root folder was created at C:\LabShares\Northstar, leaving C:\LabShares\Department and C:\LabShares\IT unchanged. In Advanced Security Settings, inherited permissions on the Northstar root were converted to explicit permissions. The two broad

AD\Users entries observed as inherited from C:\ were removed from the Northstar root only. The AD\Administrators, SYSTEM, and CREATOR OWNER entries were retained as reported.

Five folders were created under the Northstar root. Each matching departmental Global Security group was assigned Allow / Modify with scope "This folder, subfolders and files." Windows automatically selected the component permissions included in Modify; those component permissions were left enabled. Departmental users were not granted Full Control through these group entries.

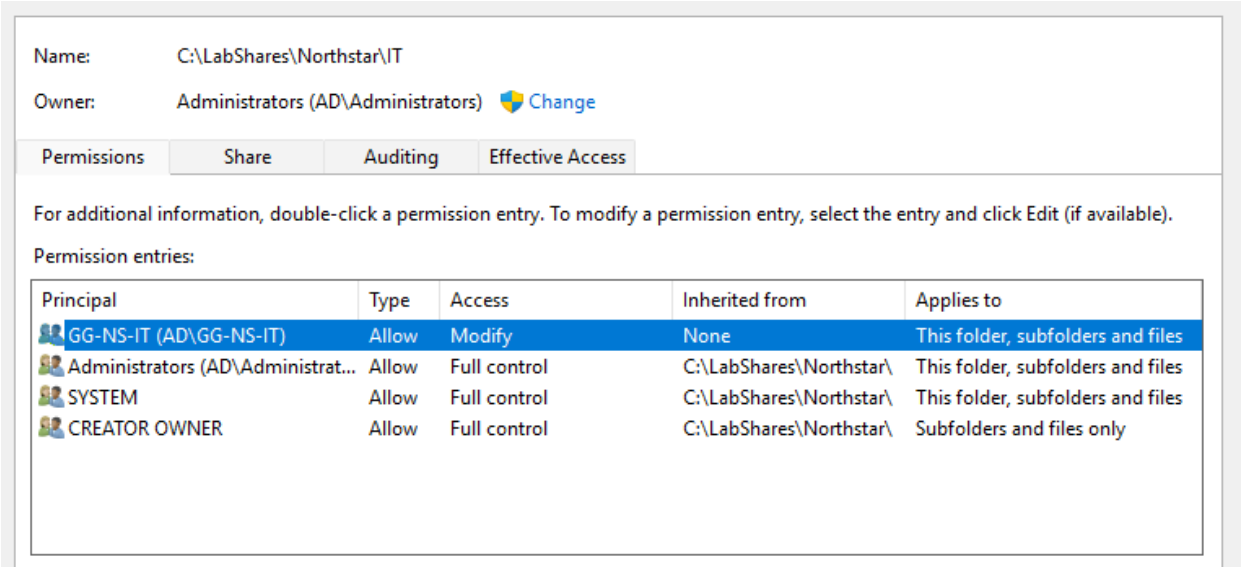


Figure 2. Advanced Security Settings for the Northstar IT folder showing GG-NS-IT with Allow / Modify applying to this folder, subfolders and files; administrative and system entries remain.

Local folder	Departmental principal	NTFS permission
C:\LabShares\Northstar\IT	GG-NS-IT	Allow - Modify; this folder, subfolders and files
C:\LabShares\Northstar\Human Resources	GG-NS-HR	Allow - Modify; this folder, subfolders and files
C:\LabShares\Northstar\Finance	GG-NS-Finance	Allow - Modify; this folder, subfolders and files
C:\LabShares\Northstar\Operations	GG-NS-Operations	Allow - Modify; this folder, subfolders and files
C:\LabShares\Northstar\Customer Support	GG-NS-CustomerSupport	Allow - Modify; this folder, subfolders and files

3.6 SMB shares and access validation

The five departmental folders were shared through Windows Advanced Sharing. The reported configuration used the matching departmental group with Change and Read at the SMB share layer and AD\Administrators with Full Control. The SMB and NTFS layers work together to determine effective access. No ACL export or screenshot of the final SMB share-permission dialog was preserved, so SMB configuration details here reflect the recorded GUI work and the user-based behavior tests below. The representative NTFS screenshot is shown in Figure 2.

Department	Actual SMB share name	User-tested UNC path
IT	Northstar-IT	\\LAB-DC01\Northstar-IT
Human Resources	Northstar-HR	\\LAB-DC01\Northstar-HR
Finance	Finance	\\LAB-DC01\Finance
Operations	Operations	\\LAB-DC01\Operations
Customer Support	Customer Support	\\LAB-DC01\Customer Support

Finance, Operations, and Customer Support were left with their shorter share names after those shares were created without the Northstar- prefix. The working share definitions were retained rather than recreated solely for naming consistency.

Department / test account	Own share	Write-test file	Selected cross-department result
IT **Jordan Blake (jordan.blake)**	\\LAB-DC01\Northstar-IT	IT-Access-Test.txt created and saved	HR share could not be opened by Jordan
Human Resources **Jamie Foster (jamie.foster)**	\\LAB-DC01\Northstar-HR	HR-Access-Test.txt created and saved	IT share could not be opened by Jamie
Finance **Alex Turner (alex.turner)**	\\LAB-DC01\Finance	Finance-Access-Test.txt created and saved	IT share could not be opened by Alex
Operations **Logan Price (logan.price)**	\\LAB-DC01\Operations	Operations-Access-Test.txt created and saved	Finance share could not be opened by Logan
Customer Support **Emerson Ward (emerson.ward)**	\\LAB-DC01\Customer Support	CustomerSupport-Access-Test.txt created and saved	IT share could not be opened by Emerson

For every department, the designated employee opened the department share and created and saved the named test file. Five selected cross-department attempts did not open the target share. The supplied evidence includes a Windows permission-denied dialog for \\LAB-DC01\Finance, but the signed-in account is not visible and the capture does not document all five attempts. The remaining outcomes were recorded during interactive testing; no specific Windows error code or additional mechanism is inferred.

Taken together, matching group membership, NTFS Modify rights, SMB permissions, and positive/negative access tests demonstrate an RBAC-style model for the tested accounts and shares. The exercise did not test every account or every possible cross-department combination.

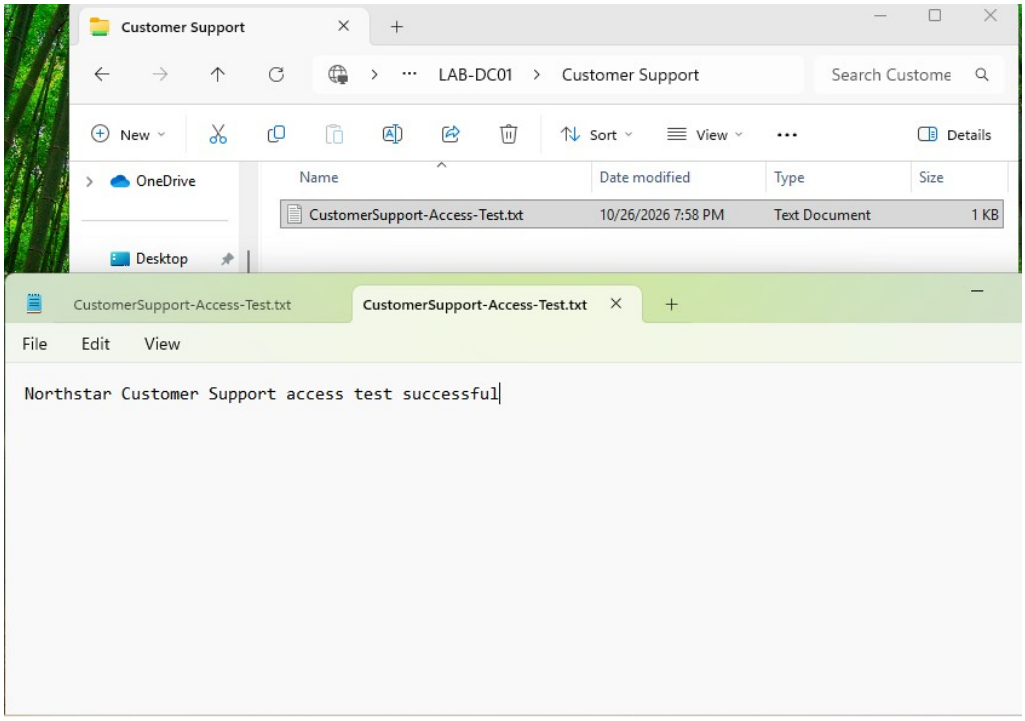


Figure 3. Successful Customer Support share write test: CustomerSupport-Access-Test.txt is present and contains the confirmation text used in the exercise.

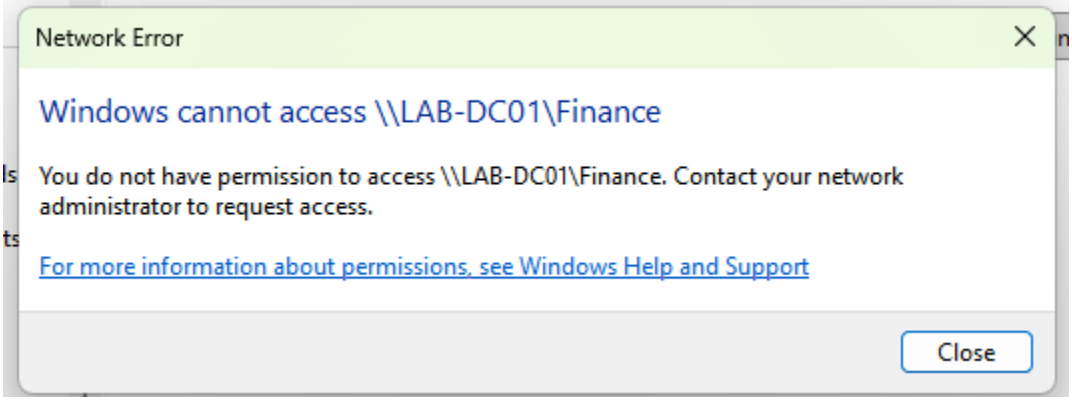


Figure 4. Windows denied access to \\LAB-DC01\Finance. The captured dialog does not identify the signed-in account, so it is supplementary evidence of a rejected access attempt.

4. Network Troubleshooting Case Study

LAB-CLIENT01 could not reach LAB-WAZUH01 - October 20, 2026

Symptom. LAB-CLIENT01 could not ping LAB-WAZUH01 at the recorded address 192.168.247.20. Windows returned "Transmit failed. General failure." The client had an APIPA address in the 169.254.x.x range, indicating that it had not obtained a usable address on the intended lab subnet.

Investigation. The client adapter was configured for custom VMnet1 host-only networking. VMware Virtual Network Editor showed that DHCP for VMnet1 was disabled. The evidence pointed to failed address assignment on the virtual network; the ping failure alone did not demonstrate that Wazuh itself was down.

Corrective action. VMnet1 DHCP was re-enabled. No Wazuh application configuration change was needed to correct the client addressing symptom.

Verification. LAB-CLIENT01 obtained 192.168.247.128/24. Ping to LAB-WAZUH01 at 192.168.247.20 then succeeded, and Wazuh agent 002 became Active.

Technical lesson Validate the adapter, virtual network, address assignment, subnet, DNS, and layer-3 reachability before investigating an application-level failure. The IP values in this case are dated observations, not a guarantee of current live state.

5. Wazuh Security Monitoring

LAB-WAZUH01 hosts the Wazuh manager, indexer, and dashboard. LAB-CLIENT01 is the documented Windows endpoint with Wazuh agent ID 002. The monitoring work focused on generating selected controlled Windows events, locating their records in Wazuh, and distinguishing a verified rule match from an attempted test without a matching result.

5.1 Verified detections and event observations

Scenario	Windows event / Wazuh evidence	Recorded result
Failed Windows logon	Security Event ID 4625; Wazuh rule 60122, "logon failure-unknown user or password."	Verified in Wazuh. A repeated attempt was supplementary evidence, not a separate scenario.
Local account creation	Security Event ID 4720 for local account WazuhLabTemp on LAB-CLIENT01; Wazuh rule 60109, "user account enabled or created."	Verified in Wazuh Discover.
Local account disabled / changed / re-enabled	Windows Security Event IDs 4725 and 4738. The user reported that the events appeared in Wazuh and screenshots were captured.	Reported as verified; exact timestamps, complete event fields, and screenshot filenames were not retained in the available record.
Account-lockout and recovery exercise	The user reported completing the exercise. Exact event IDs, Wazuh rule details, timestamps, and screenshot details were not available.	Completed per user confirmation; no specific event/rule is claimed.

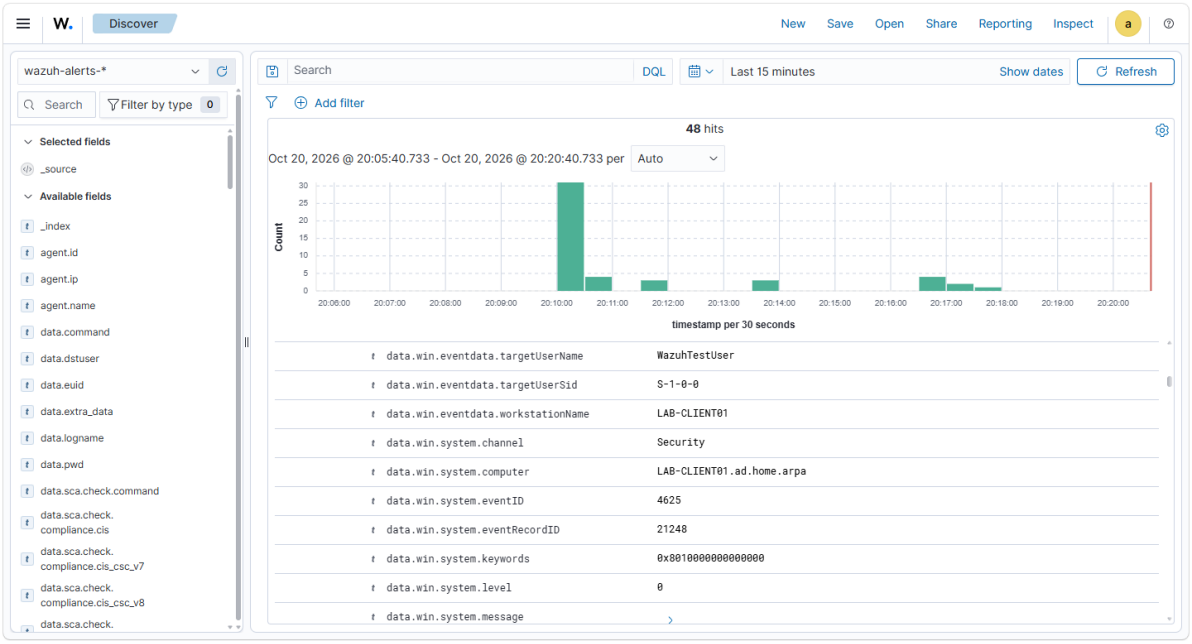


Figure 5. Wazuh Discover event details showing Windows Security Event ID 4625 for LAB-CLIENT01. The recorded rule 60122 match is summarized in the verified-detection table.

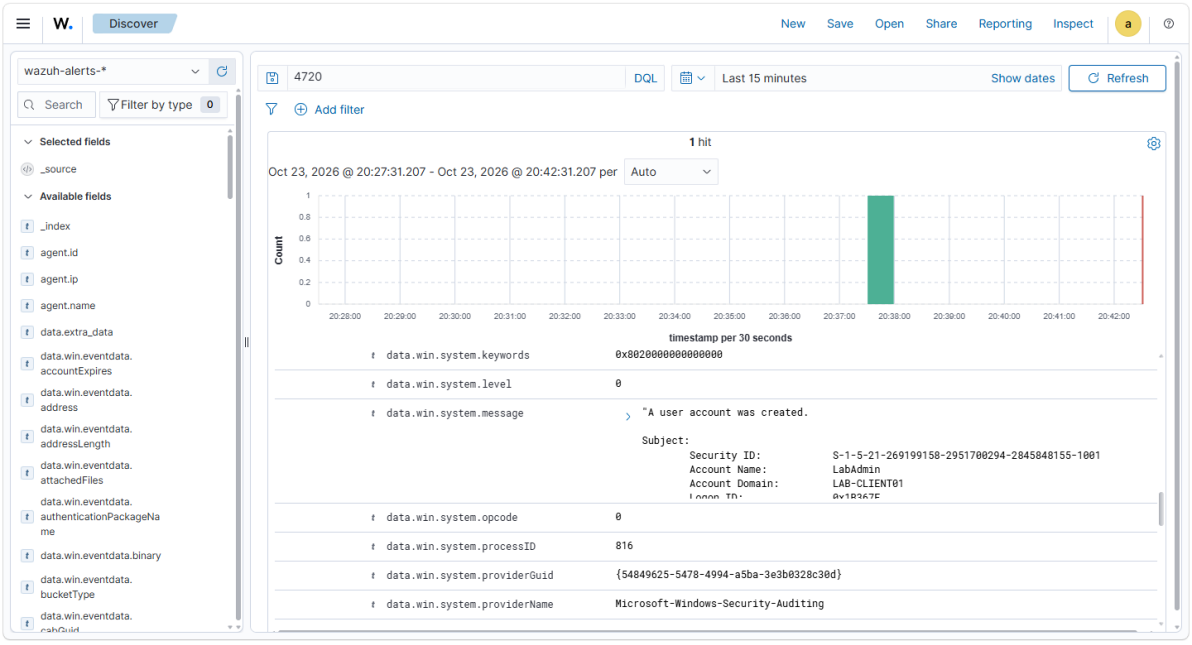


Figure 6. Wazuh Discover view filtered for Event ID 4720, with the account-creation event details visible. The recorded rule 60109 match is summarized in the verified-detection table.

5.2 Attempts where detection was not verified

Scenario	What was checked	Conclusion
AD group membership: GG-Wazuh-Test / wazuh.test	Expected event/alert was not located in the checked Windows log or Wazuh. The relevant change attempt included LAB-DC01, while the known active agent in the record was on LAB-CLIENT01.	Detection not verified. The precise reason for the missing event was not established.
Print Spooler service stop/start	Expected Windows System Event ID 7036 was not found in the checked log or Wazuh.	Wazuh detection not verified. This is separate from GLPI Ticket 7, where the service state change and recovery were verified directly.

A Windows DNS Client Event ID 1014 was visible in Wazuh Discover at an earlier checkpoint. Its presence demonstrates that an event record was searchable at that time; it does not by itself prove that a dedicated security detection or response workflow was successful.

Wazuh deployment required troubleshooting. Later project records show the manager, indexer, and dashboard operational at a checkpoint, but an earlier discrepancy in multi-service status output was not fully diagnosed. The available record does not establish a specific Wazuh alert linked to any GLPI ticket, and no Wazuh detection is claimed for the Northstar account, group, share, or file-access changes.

6. GLPI Service Desk and Ticket Register

GLPI on LAB-LNX01 is used to document the service-desk side of the lab: user-reported symptoms, affected systems, impact, investigation, corrective actions, verification, and ticket closure. The combined GLPI export provides the original details for all 12 tickets. In the export, all 12 tickets are marked Closed and show Solution Approved; the technical meaning of that administrative status is considered alongside the recorded verification.

The following ticket entries retain the actual ticket IDs and titles from the export. Dates and times are shown as recorded by GLPI; the export did not state a timezone. All tickets are recorded as Incident type with Helpdesk request source and Medium urgency/impact/priority.

Ticket 01 | LAB-CLIENT01 - User unable to sign in after account lockout

Requester: Alice Carter | **Asset:** LAB-CLIENT01 | **Status:** Closed / Solution Approved

Opened: 2026-10-14 03:48:53 | **Resolved:** 2026-10-15 01:50:26 | **Closed:** 2026-10-15 01:51:15

Issue and impact. One user could not sign in to a Windows workstation and suspected the account was locked following repeated unsuccessful attempts.

Investigation and corrective action. The lab domain's account-lockout threshold was temporarily set to three attempts to reproduce the reported condition. Alice Carter's Active Directory account locked after three unsuccessful sign-in attempts. The account was unlocked through Active Directory Users and Computers on LAB-DC01. After the controlled test, the domain's original threshold value of 0 was restored.

Verification and outcome. Alice signed in to LAB-CLIENT01 using her existing domain credentials. The domain lockout threshold was checked and confirmed restored to 0.

Scope or verification boundary. The recorded threshold value of 0 means the domain password policy does not enforce account lockout via that threshold. This was the original lab setting restored after reproduction; it should not be represented as a production security recommendation.

Ticket 02 | LAB-CLIENT01 - Bob Martinez required to change password before sign-in

Requester: Bob Martinez | **Asset:** LAB-CLIENT01 | **Status:** Closed / Solution Approved

Opened: 2026-10-15 02:12:46 | **Resolved:** 2026-10-15 02:21:24 | **Closed:** 2026-10-15 02:21:45

Issue and impact. Bob could not complete sign-in because Windows required a password change. His first replacement-password attempt was rejected for not meeting the domain complexity requirements.

Investigation and corrective action. The password-change prompt was reproduced at Windows sign-in. The initial replacement password failed the domain complexity requirements. Bob selected a different password that met the requirements and completed the change at the sign-in screen.

Verification and outcome. Bob successfully signed in to LAB-CLIENT01 with the new password. User access was restored; no password value is included in this portfolio.

Ticket 03 | LAB-CLIENT01 - Bob Martinez unable to modify Department shared file

Requester: Bob Martinez | **Asset:** LAB-CLIENT01 | **Status:** Closed / Solution Approved

Opened: 2026-10-15 03:11:07 | **Resolved:** 2026-10-15 03:14:24 | **Closed:** 2026-10-15 03:14:26

Issue and impact. Bob could open the file \\LAB-DC01\Department\AccessTest.txt but received a permission error when trying to save changes. The impact was read access without modify access.

Investigation and corrective action. The Department share's NTFS permissions granted Modify to GG-Lab-Users while the built-in Users group had read-level permissions. Bob had been temporarily removed from GG-Lab-Users for controlled testing. His membership was restored, and he signed out and back in to refresh his Windows security token.

Verification and outcome. Bob successfully edited and saved AccessTest.txt after signing back in. Modify access was verified and the issue was resolved.

Ticket 04 | LAB-CLIENT01 - DNS name resolution failure after incorrect DNS configuration

Requester: Bob Martinez | **Asset:** LAB-CLIENT01 | **Status:** Closed / Solution Approved

Opened: 2026-10-15 03:37:47 | **Resolved:** 2026-10-15 03:41:24 | **Closed:** 2026-10-15 03:41:26

Issue and impact. LAB-CLIENT01 could not resolve the LAB-DC01 hostname. DNS queries timed out while the client was configured to use an incorrect DNS server address.

Investigation and corrective action. The client was found to be using 192.168.247.99 as its DNS server. Queries to that address timed out. On LAB-DC01, the DNS service was running and a local query successfully resolved LAB-DC01 to 192.168.247.2. The preferred DNS server on LAB-CLIENT01 was restored to 192.168.247.2.

Verification and outcome. From LAB-CLIENT01, a direct DNS query to 192.168.247.2 and a normal hostname lookup for LAB-DC01 both succeeded. The incorrect setting had been introduced deliberately for the controlled exercise.

Ticket 05 | LAB-LNX01 - GLPI unavailable due to stopped Apache service

Requester: Alice Carter | **Asset:** LAB-LNX01 | **Status:** Closed / Solution Approved

Opened: 2026-10-15 03:51:49 | **Resolved:** 2026-10-15 03:54:51 | **Closed:** 2026-10-15 03:54:53

Issue and impact. The GLPI login page at http://192.168.247.10 could not be reached from LAB-CLIENT01, making the lab help-desk application temporarily unavailable.

Investigation and corrective action. As a controlled service-failure exercise, Apache on LAB-LNX01 was stopped. While the service was stopped, the GLPI login page was unreachable. Apache was restarted using sudo systemctl start apache2 (with service state checked via systemctl).

Verification and outcome. Apache returned to active (running), and the GLPI login page loaded successfully from LAB-CLIENT01. The service interruption was introduced deliberately for the exercise.

Scope or verification boundary. This demonstrates controlled service interruption and recovery. It does not establish that an unexpected, pre-existing Apache failure caused a real outage.

Ticket 06 | LAB-CLIENT01 - Access denied to IT network share

Requester: Alice Carter | **Asset:** LAB-CLIENT01 | **Status:** Closed / Solution Approved

Opened: 2026-10-15 04:18:57 | **Resolved:** 2026-10-15 04:19:57 | **Closed:** 2026-10-15 04:19:59

Issue and impact. Access to the IT network share \\LAB-DC01\IT was denied from LAB-CLIENT01 while the user was signed in with the local workstation account LAB-CLIENT01\LabAdmin.

Investigation and corrective action. Investigation showed that access to the IT folder was restricted through the GG-Lab-IT security group, which contained Alice Carter. The share-level Everyone permission did not override the restrictive NTFS permissions. No share or NTFS permissions were changed during this investigation.

Verification and outcome. The user signed out of the local workstation account and signed in with Alice Carter's domain account. Alice successfully opened \\LAB-DC01\IT with her domain credentials. The existing folder restrictions were preserved.

Ticket 07 | LAB-CLIENT01 - Simulated printing failure caused by stopped Print Spooler

Requester: Alice Carter | **Asset:** LAB-CLIENT01 | **Status:** Closed / Solution Approved

Opened: 2026-10-16 01:52:46 | **Resolved:** 2026-10-16 01:53:48 | **Closed:** 2026-10-16 01:53:50

Issue and impact. The simulated user symptom was that printing was unavailable from LAB-CLIENT01.

Investigation and corrective action. The Print Spooler service was initially Running. For the controlled exercise, the service was deliberately stopped and its status was confirmed as Stopped. It was then restarted using PowerShell with Start-Service Spooler.

Verification and outcome. Get-Service Spooler showed the service had returned to Running. The service interruption and recovery were directly verified.

Scope or verification boundary. No printer was installed on the virtual workstation, so an actual print job could not be tested. The printing complaint was simulated; the underlying service state change was real and verified.

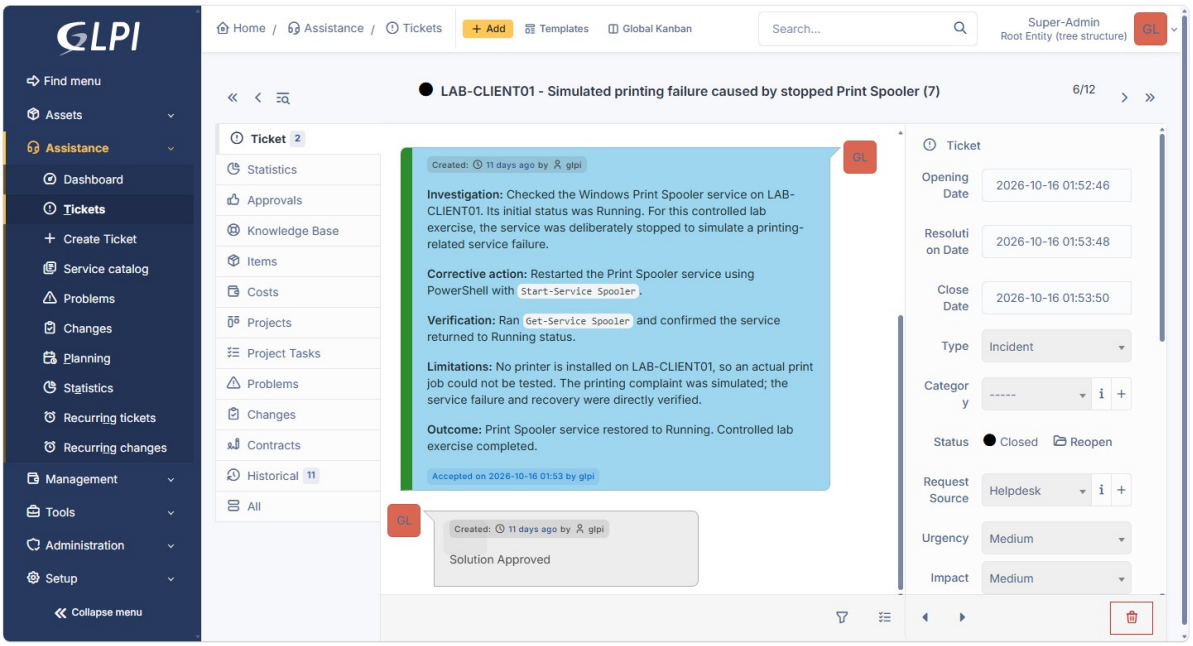


Figure 7. GLPI Ticket 7 records the controlled Print Spooler stop/restart, Running-state verification, and the limitation that no print job could be tested because no printer was installed.

Ticket 08 | LAB-CLIENT01 - Controlled BITS service interruption and recovery

Requester: Bob Martinez | **Asset:** LAB-CLIENT01 | **Status:** Closed / Solution Approved

Opened: 2026-10-16 02:01:44 | **Resolved:** 2026-10-16 02:02:21 | **Closed:** 2026-10-16 02:02:22

Issue and impact. A controlled help-desk exercise was used to practice troubleshooting and recovery of the Background Intelligent Transfer Service (BITS) on LAB-CLIENT01.

Investigation and corrective action. PowerShell inspection showed BITS was initially Running. The service was deliberately stopped, its Stopped state was checked, and it was restarted with Start-Service BITS.

Verification and outcome. Get-Service BITS confirmed that the service returned to Running. Service recovery was verified.

Ticket 09 | LAB-DC01 - Active Directory test account disabled and restored

Requester: Bob Martinez | **Asset:** LAB-DC01 | **Status:** Closed / Solution Approved

Opened: 2026-10-16 02:13:38 | **Resolved:** 2026-10-16 02:14:20 | **Closed:** 2026-10-16 02:14:21

Issue and impact. This was a controlled Active Directory account-lifecycle administration exercise using a dedicated test account, not a report of an actual employee outage.

Investigation and corrective action. Using Active Directory Users and Computers on LAB-DC01, the dedicated domain account test.user (display name Test User) was created in the domain Users container. The account was checked as enabled, deliberately disabled, and then re-enabled.

Verification and outcome. The account properties were reopened and the Account is disabled option was confirmed unchecked. The test account was restored to an enabled state; no existing user account was modified.

Ticket 10 | LAB-CLIENT01 - Temporary Group Policy refresh failure

Requester: Alice Carter | **Asset:** LAB-CLIENT01 | **Status:** Closed / Solution Approved

Opened: 2026-10-16 02:59:25 | **Resolved:** 2026-10-16 03:00:42 | **Closed:** 2026-10-16 03:00:43

Issue and impact. A forced computer Group Policy refresh on the domain-joined workstation initially failed with a message indicating a lack of network connectivity to a domain controller.

Investigation and corrective action. The computer-side Group Policy results listed LAB-Workstation-Baseline, LAB-Workstation-Security, and Default Domain Policy. A ping to LAB-DC01 returned four replies, and nltest /dsgetdc:ad.home.arpa successfully returned domain-controller information. The computer policy refresh was retried with gpupdate /target:computer /force.

Verification and outcome. The retry completed successfully. No DNS, network, or Group Policy settings were changed during the exercise.

Scope or verification boundary. The initial failure was observed and the retry succeeded, but the cause of the first failure was not determined. The ticket does not establish that a particular network, DNS, or policy fault was repaired.

Ticket 11 | LAB-CLIENT01 - Investigate Group Policy-managed Windows Firewall rule

Requester: Bob Martinez | **Asset:** LAB-CLIENT01 | **Status:** Closed / Solution Approved

Opened: 2026-10-16 03:12:58 | **Resolved:** 2026-10-16 03:13:53 | **Closed:** 2026-10-16 03:13:55

Issue and impact. This controlled troubleshooting exercise investigated a Group Policy-managed Windows Firewall rule and checked workstation connectivity.

Investigation and corrective action. The Domain, Private, and Public firewall profiles were enabled. Ethernet0 was categorized as DomainAuthenticated. The enabled inbound rule LAB-Allow ICMPv4 Echo Requests was scoped to the Domain profile and identified as sourced from Group Policy. Inspection confirmed ICMPv4 with no port restrictions. An attempt to manage the rule locally by display name did not find a matching locally manageable rule, consistent with the rule being policy-managed.

Verification and outcome. From LAB-CLIENT01, ping to LAB-DC01 at 192.168.247.2 returned four replies. No firewall or Group Policy settings were changed, and no outage was reproduced. The investigation verified the rule source and connectivity during the test.

Scope or verification boundary. This was a policy-inspection and connectivity investigation, not a demonstrated firewall repair.

Ticket 12 | LAB-LNX01 - Linux test user denied access to file

Requester: Alice Carter | **Asset:** LAB-LNX01 | **Status:** Closed / Solution Approved

Opened: 2026-10-16 03:44:30 | **Resolved:** 2026-10-16 03:45:25 | **Closed:** 2026-10-16 03:45:26

Issue and impact. A controlled Linux permissions exercise reproduced a Permission denied result when the separate labtest account attempted to read /home/mike/LabAccessTest/access-test.txt. The file initially allowed the owner to read and write while other users had read-only access; the parent directory /home/mike had permissions 750.

Investigation and corrective action. The file mode was changed to 600 for reproduction, and labtest received Permission denied. A labaccess group was created and both mike and labtest were added to it. The test directory and file group ownership and permissions were changed. Inspection showed that /home/mike remained restricted, preventing labtest from traversing the complete path.

Verification and outcome. Group membership was checked and confirmed for labtest, including labaccess as a supplementary group. Successful post-change file access was not verified.

Scope or verification boundary. The GLPI record is administratively closed and marked Solution Approved, but the technical remediation remains unverified because the parent-directory traversal restriction remained. The portfolio does not claim successful recovery.

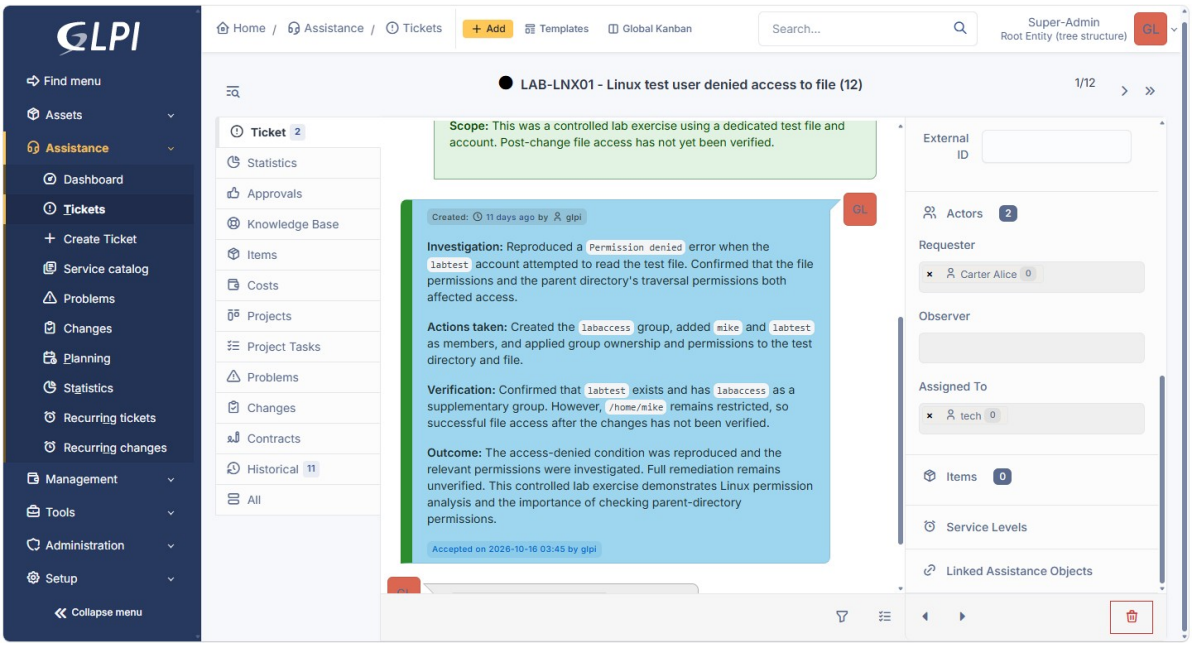


Figure 8. GLPI Ticket 12 records the permission investigation and explicitly notes that successful post-change access was not verified because /home/mike remained restricted.

7. Snapshots and Recovery Practice

VMware snapshots were used as convenient rollback checkpoints before or after significant lab milestones. Snapshot presence is documented where it was specifically reported or verified. Snapshots are not independent backups and do not replace a separate backup strategy for important data.

VM / checkpoint	Recorded scope and status
LAB-DC01, LAB-CLIENT01, LAB-WAZUH01, LAB-LNX01 - Post-Wazuh-Exercise-2026-10-25	Reported taken while each VM was powered off and verified in VMware Snapshot Manager on October 25, 2026.
LAB-DC01 - Post-Northstar-AD-Users-Groups-2026-10-26	Taken while powered off after the Northstar OU hierarchy, 25 user accounts, five departmental Global Security groups, and memberships were confirmed. Snapshot Manager verification was reported. Restore/revert was not tested.
LAB-DC01 - Final Northstar Configuration	User reported creating this snapshot on October 26 after the Northstar departmental folders, NTFS permissions, SMB shares, and the five-department access tests. An independent Snapshot Manager screenshot was not included in the record.

8. Verification Boundaries and Technical Limitations

The lab record intentionally distinguishes the outcome that was demonstrated from the scope that remains untested. The following limitations are material to interpreting the portfolio:

- Network addresses and host/network settings are historical checkpoints. Current configuration should be verified before relying on an earlier value.
- Northstar access testing used one designated account per department. Each account successfully opened and wrote to its own share, and five selected cross-department attempts did not open the target share; not every user or every possible share combination was tested.
- The supplied screenshot set includes a Windows access-denied dialog for \\LAB-DC01\Finance, but the signed-in username is not visible in that capture. This version also includes one representative NTFS permissions screenshot showing GG-NS-IT with Allow / Modify on the Northstar IT folder, subfolders, and files. A screenshot of the final SMB share-permission dialog was not included. The documented permission assignments and five-department test outcomes are recorded from the interactive exercise; no specific Windows error code is inferred.
- No Wazuh detection is claimed for Northstar user creation, group membership, SMB access, or file writes. Wazuh-to-GLPI linkage is not established by the available evidence.
- Ticket 5 is a controlled Apache service interruption and recovery. Ticket 7 confirms Print Spooler service-state recovery but not successful printing because no printer was installed.
- Ticket 10 resolved on retry, but the root cause of the initial Group Policy refresh failure was not established.
- Ticket 11 verified the Group Policy source of the ICMPv4 rule and successful ping connectivity during investigation; it did not change or repair a firewall policy.
- Ticket 12 remains technically unverified: the test user could not traverse the full path because /home/mike remained restricted after the other permission changes.
- Wazuh failed-logon and local-account-related detections are documented where verified. The AD group-membership and Print Spooler event-detection attempts remain unverified.

Scope statement This portfolio documents selected, hands-on lab outcomes. It does not claim production enterprise operation, exhaustive security validation, comprehensive disaster-recovery testing, or a fully verified alert-to-ticket response integration.

Appendix A. Key Technical Reference

Reference	Recorded value
Active Directory domain	ad.home.arpa
Domain controller	LAB-DC01; historical address 192.168.247.2
Windows endpoint	LAB-CLIENT01; Wazuh agent ID 002; 192.168.247.128/24 observed during October 20 DHCP recovery; earlier value 192.168.247.101 also recorded
Wazuh endpoint	LAB-WAZUH01 at 192.168.247.20 (recorded)
Linux / GLPI endpoint	LAB-LNX01 at 192.168.247.10 (recorded)
VMnet1	Host-only; 192.168.247.0/24; host adapter 192.168.247.1; DHCP server 192.168.247.254 when enabled
VMnet8	NAT; 192.168.2.0/24 (recorded)
Existing shares	Department -> C:\LabShares\Department; IT -> C:\LabShares\IT
Northstar shares	Northstar-IT; Northstar-HR; Finance; Operations; Customer Support
Verified Wazuh example	Windows Security Event ID 4625 -> Wazuh rule 60122
Verified Wazuh example	Windows Security Event ID 4720 -> Wazuh rule 60109
Additional reported Wazuh detections	Windows Security Event IDs 4725 and 4738 for local account disable/change/re-enable scenario